DroneVault – A Secure Drone Operations & Analytics Platform

VJTI

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PROBLEM STATEMENT:

The rapid growth of consumer drones has created two major challenges: lack of a private, offline tool to analyze flight data, and increasing security risks from unauthorized drones. Existing solutions depend on cloud services, raising data privacy concerns, while traditional counter-drone methods disrupt nearby networks. Therefore, there is a need for a secure, platform that enables efficient drone data analysis and safe monitoring without relying on external systems.

ABSTRACT:

DroneVault is a privacy-first platform designed for drone forensics and security. It allows the users to import and analyze DJI and Litchi flight logs, store telemetry data in a local database, and visualize flights through interactive 3D replays, charts, and battery health reports. Our platform also supports exporting data for reporting and compliance. In addition, it includes a security module that monitors nearby Wi-Fi networks, identifies potential rogue drone threats, and selectively blocks the unauthorized drones using deauthentication without affecting trusted networks.

APPLICATIONS:

- **Security & Surveillance** – Detect unauthorized drones in restricted areas.
- **Critical Infrastructure Protection** – Prevent UAV threats near sensitive facilities like power grids.
- **Fleet Management** – Track logs, battery health, and for UAV maintenance.
- **Incident Analysis** – Replay 3D flights for investigation.
- **Academic Research** – Study Wi-Fi protocols, drone failsafe behaviours, and battery degradation patterns.

Feature	DroneVault	Cloud-based Logbooks	Flight Radar
Data Privacy and Storage	✔ 100% Local	✘ Cloud Upload	✔ 100% Local or Cloud
Security	✔ Detects & Blocks Rogue Drones	✘ None	✘ None
Flight Replay	✔ Interactive 3D	✘ 2D Only	~ Limited 3D
Report Generation	✔ Yes	✘ No	✘ No
Price	✔ Open Source Project	✘ $3.99 / month	✘ $10 / month

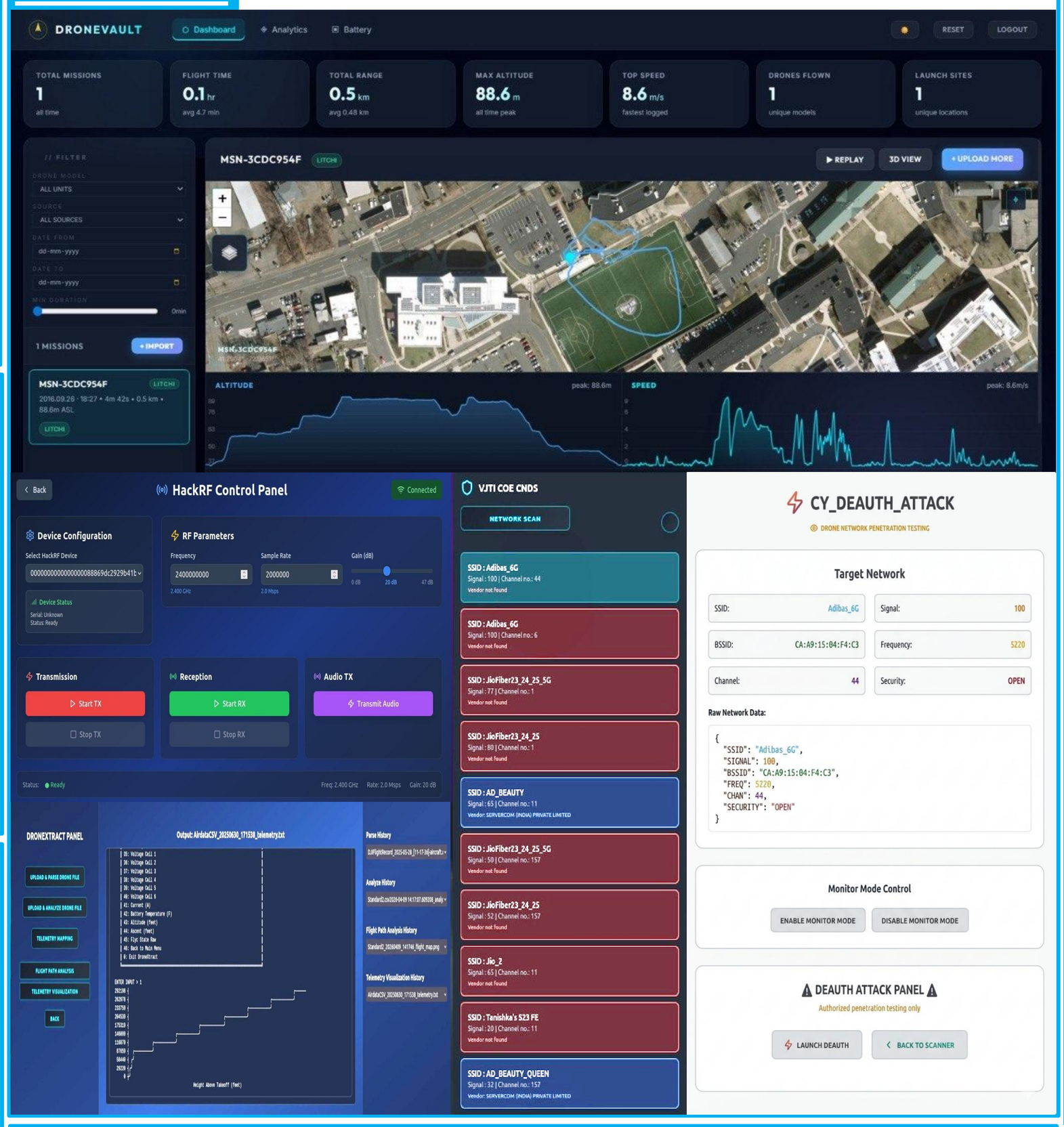
CONCLUSIONS:

Our project combines drone forensics and security in a local platform. It analyzes flight logs and detects unauthorized drones while keeping all data private. Built with modern technologies, it is a scalable and deployable solution for real-world use.

Future Scope:

AI/ML-based forensic analysis, and advanced UAVs threat detection and mitigation.

RESULTS:



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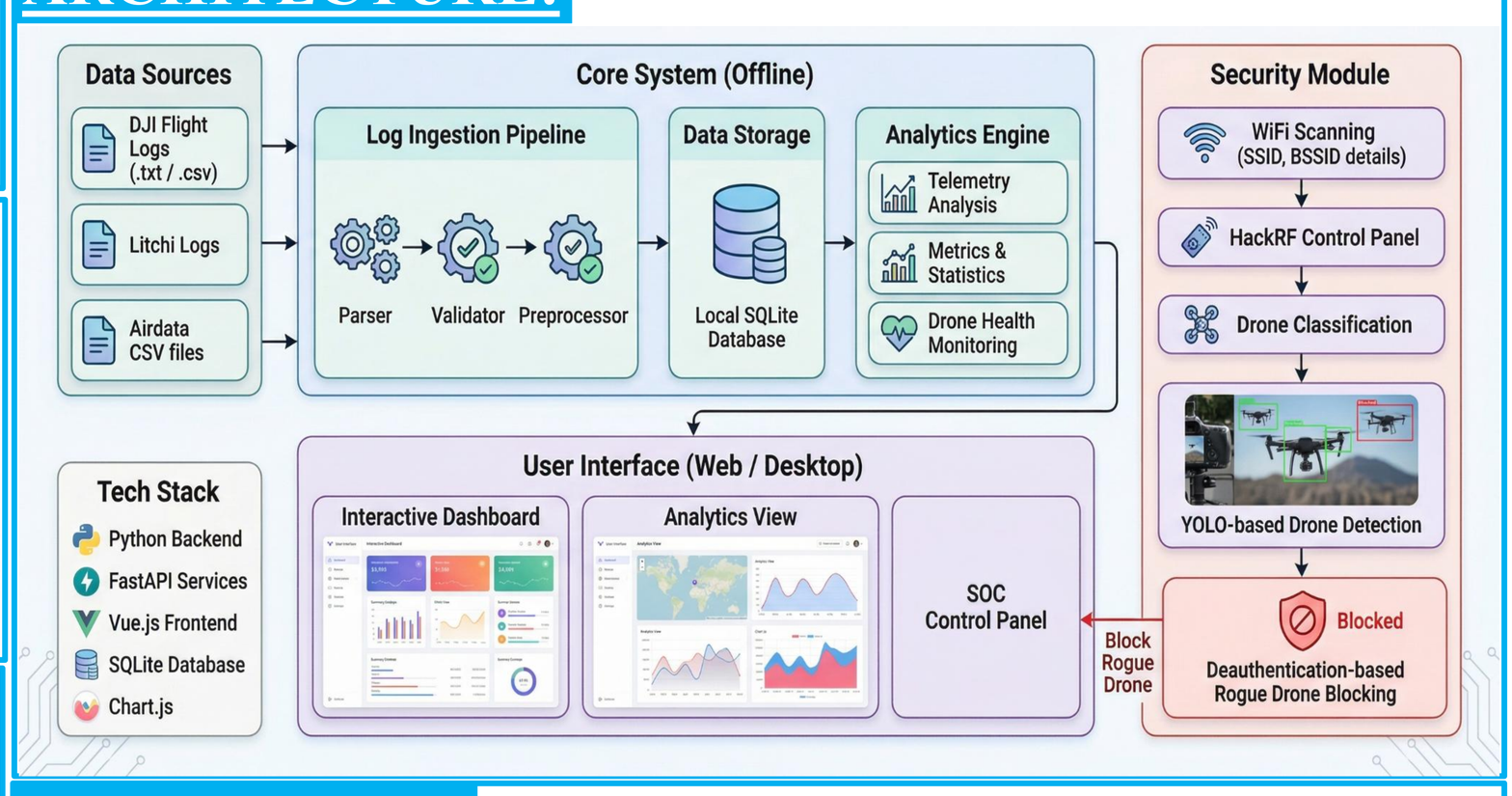

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